

IMPLEMENTATION OF INTEGRITY CONSTRAINTS

ASSIGNMENT 6

DATABASE MANAGEMENT SYSTEM

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INTRODUCTION

Integrity constraints maintain accuracy, validity, and consistency in database tables. This assignment implements primary key, foreign key, UNIQUE, CHECK, and DEFAULT constraints in the Online Shopping Management System using MySQL.

1. PRIMARY KEY AND FOREIGN KEY CONSTRAINTS

Primary keys uniquely identify records, while foreign keys establish relationships and maintain referential integrity.

Table	Primary Key	Foreign Key
CUSTOMER	customer_id	—
CATEGORY	category_id	—
SUPPLIER	supplier_id	—
PRODUCT	product_id	category_id, supplier_id
ORDERS	order_id	customer_id
ORDER_ITEM	order_item_id	order_id, product_id
PAYMENT	payment_id	order_id
DELIVERY	delivery_id	order_id

Primary keys prevent duplicate identifiers and NULL primary key values. Foreign keys ensure that related records refer to existing parent records.

2. IMPLEMENTING UNIQUE, CHECK, AND DEFAULT CONSTRAINTS

A. UNIQUE CONSTRAINT

UNIQUE prevents duplicate values. Customer and supplier email addresses can be made unique, and an order can be restricted to one payment and one delivery record.

```
USE Online_Shopping_DB;

ALTER TABLE CUSTOMER
ADD CONSTRAINT uq_customer_email UNIQUE (email);

ALTER TABLE SUPPLIER
ADD CONSTRAINT uq_supplier_email UNIQUE (email);

ALTER TABLE PAYMENT
ADD CONSTRAINT uq_payment_order UNIQUE (order_id);

ALTER TABLE DELIVERY
ADD CONSTRAINT uq_delivery_order UNIQUE (order_id);
```

B. CHECK CONSTRAINT

CHECK restricts values stored in a column.

```
ALTER TABLE PRODUCT
ADD CONSTRAINT chk_product_price CHECK (price >= 0);

ALTER TABLE PRODUCT
ADD CONSTRAINT chk_product_stock CHECK (stock_quantity >= 0);

ALTER TABLE ORDER_ITEM
ADD CONSTRAINT chk_order_item_quantity CHECK (quantity > 0);

ALTER TABLE ORDER_ITEM
ADD CONSTRAINT chk_order_item_price CHECK (unit_price >= 0);

ALTER TABLE ORDERS
ADD CONSTRAINT chk_order_total CHECK (total_amount >= 0);

ALTER TABLE PAYMENT
ADD CONSTRAINT chk_payment_amount CHECK (amount >= 0);
```

- Product prices cannot be negative.
- Stock quantities cannot be negative.
- Order-item quantity must be greater than zero.
- Order totals, unit prices, and payment amounts cannot be negative.

C. DEFAULT CONSTRAINT

DEFAULT assigns a value when an INSERT statement omits that column. The following statements use MySQL 8.0.13 or later syntax.

```
ALTER TABLE PRODUCT
ALTER COLUMN stock_quantity SET DEFAULT 0;

ALTER TABLE ORDERS
ALTER COLUMN order_status SET DEFAULT 'Pending';

ALTER TABLE PAYMENT
ALTER COLUMN payment_status SET DEFAULT 'Pending';

ALTER TABLE DELIVERY
ALTER COLUMN delivery_status SET DEFAULT 'Preparing';
```

3. TESTING THE CONSTRAINTS

```
SHOW CREATE TABLE CUSTOMER;
SHOW CREATE TABLE PRODUCT;
SHOW CREATE TABLE ORDERS;
SHOW CREATE TABLE ORDER_ITEM;
SHOW CREATE TABLE PAYMENT;
```

```
SHOW CREATE TABLE DELIVERY;
SHOW COLUMNS FROM PRODUCT;
```

The output displays the table definitions and their constraints. The Default column in SHOW COLUMNS displays the configured default value. An attempt to insert a negative price or duplicate email should be rejected by the corresponding constraint.

CONCLUSION

Primary keys identify records, foreign keys maintain relationships, UNIQUE constraints prevent duplicate values, CHECK constraints reject invalid values, and DEFAULT constraints provide predefined values. Together, these constraints improve the reliability of the Online Shopping Management System.